
NIH SBIR Inquiry: A Small-Business Path to a Governed Surgical Advisory System for the Robotic Whipple

Technical objective.

An on-premises advisory system, verified stop latency, and an audit record, taken through a Phase 1 first-in-human robotic pancreaticoduodenectomy with perioperative daraxonrasib (RMC-6236).

Commercial case.

The deliverable is a portable governance package, not a robot: a logging schema, a bench method, and a regulatory dossier that any current qualified pancreatic surgery program reaches full adoption.

CEO Kevin Kawchak , ChemicalQDevice, a small business, kevink@chemicalqdevice.com, 0009-0007-5457-8667 San Diego, California | August 7, 2026 | Intended site: UC San Diego Moores Cancer Center (feasibility stage)

Submitted as an independent scientist under the funding approach set out in Science: A New Golden Age (July 2026). Not medical or regulatory advice, and not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. UC San Diego is named as the intended partner of choice and is not a sponsor, site, or endorser.

1 A Venture That Did Not Start With a Dissertation

Science: A New Golden Age asks the roughly **\$200 billion** annual federal research portfolio to stop deferring to “the same incumbents that consume the funding” and to open pathways “grounded in demonstrated skill rather than academic pedigree alone,” naming SBIR as a program that “opens doors for technician-founded ventures, directing resources toward ideas that do not always start with a dissertation” [1].

ChemicalQDevice is that kind of venture, shown in Table 1 and Figure 1. Between June 2025 and July 2026 it produced, with one operator, a Phase 1 protocol, a Phase 2 protocol, an Investigational New Drug application, two funding applications, five legislative drafts, and four guidance works, all publicly deposited [2]. The commercial product is not the robot and not the drug. It is the governance package that makes an on-premises model usable in a regulated operating room: a logging schema, a bench stop-latency method, and a regulatory dossier a site can adopt without rebuilding either.

SBIR criterion	Status today	What the award changes
Technical merit and innovation	Protocol and IND published; three audited simulations	Moves the claim from in silico to first-in-human
Qualification of the small business	One operator, 14 relevant works, no indirect cost	Adds a contracted clinical site and a monitor
Commercial potential	Governance package is site-portable by construction	Produces the first adopting site and a reference deployment

Table 1. The three standing SBIR review criteria, the status of each today, and the specific change an award produces. Nothing in the middle column depends on the award having been made.

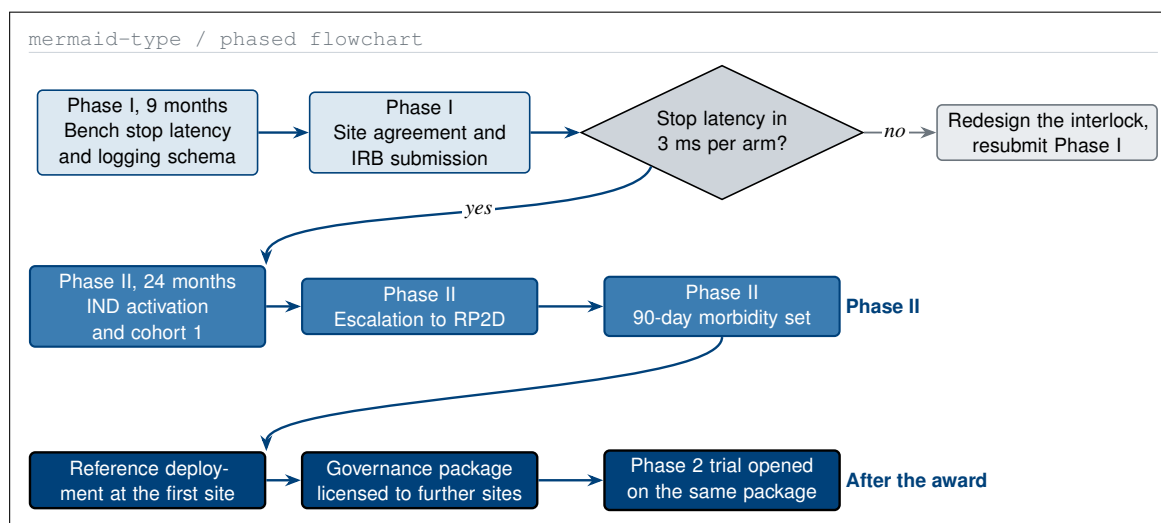


Figure 1. The Phase I, Phase II, and post-award pathway. The single gate between Phase I and Phase II is a measured number, not a review, and a failure there returns to Phase I rather than ending the venture.

2 What the Company Has Already Demonstrated

The strongest commercial evidence a pre-award small business can offer is its own unit economics. Industry benchmarks put a single virtual trial run above \$120,000 and 4.5 months (Table 2), and a quantitative systems pharmacology virtual trial above \$2 million and several months to a year or more. The three simulations here cost a projected \$36,330 each and took about a month each [3].

Work product	Industry benchmark	This company
Empirical virtual trial, triplicate	> \$120,000, 4.5 months/run	\$36,330 projected, about one month
Ten-arm QSP virtual trial	> \$2,000,000, months/year	\$36,330 projected, about one month
Digital-twin simulator platform	\$28,000-\$700,000, 3-16 weeks	\$36,330 projected, about one month

Table 2. Three work products against their industry cost and schedule benchmarks. The comparison is the company's demonstrated burn rate, which is what a Phase I reviewer is actually being asked to underwrite.

On the clinical side, the August 2025 quantitative systems pharmacology run returned median overall survival of **12.8 months** for the daraxonrasib combination against **5.4** for chemotherapy [4], and RASolute 302 reported **13.2** against **6.6 months** in the RAS G12 population nine months later [5]. That proximity is chronology, not validation: 1000 simulated against 241 enrolled, a combination against a single agent, KRAS G12C against a primarily G12D and G12V population.

3 The Technical Risk a Phase I Must Retire

The operation itself is a staged eight-arm robotic pancreaticoduodenectomy with perioperative daraxonrasib in resectable or borderline-resectable KRAS G12-mutated PDAC: open-label, single-arm, 3+3 escalation, up to 18 treated participants, 30-day and 90-day safety, and 24-month OS follow-up, shown in Figure 2 [6, 7].

Positioning note. First in human refers to the integrated surgical and advisory workflow, not to daraxonrasib, which is investigational and already in Phase 3 evaluation. The robotic configuration is specified by manufacturer, model, cleared configuration, arm count, and software version at the site agreement, and no drug supply or letter of authorization is in place with the agent's developer.

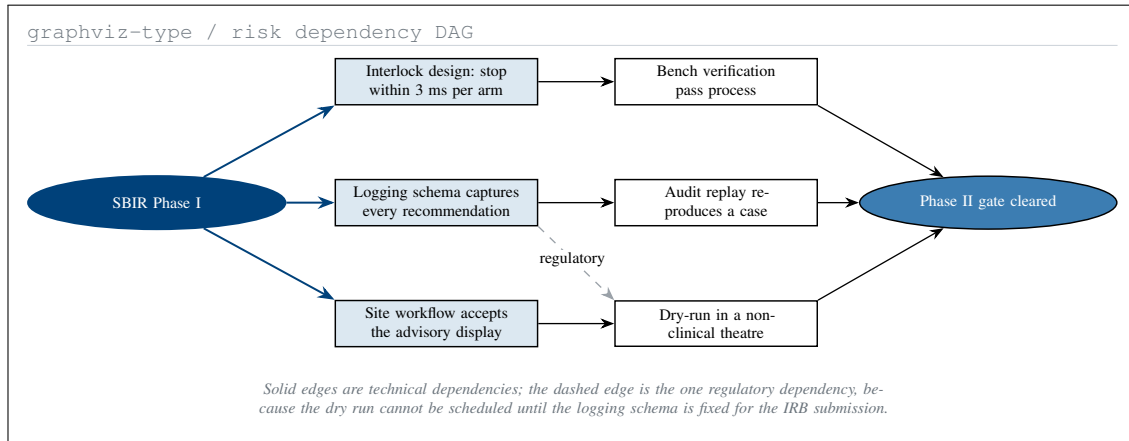


Figure 2. The three technical risks a Phase I retires, each with its own verification, and the single regulatory dependency that couples the logging schema to the scheduling of the dry run.

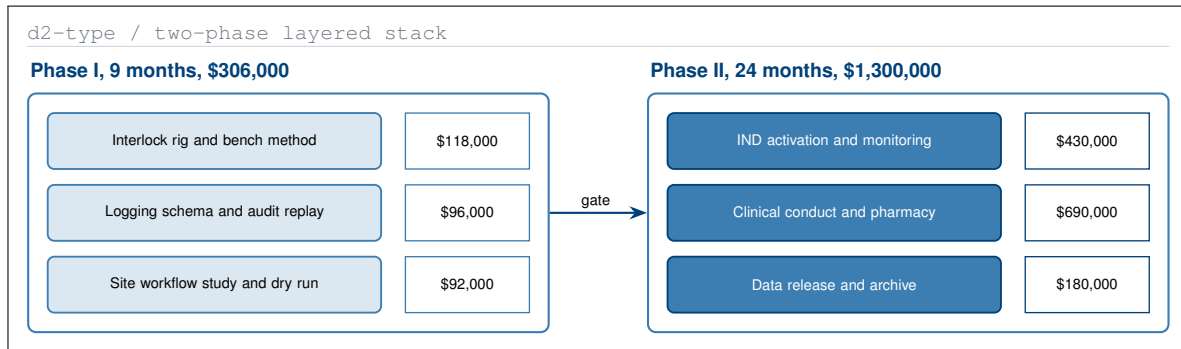


Figure 3. Phase I and Phase II budgets by layer, with the measured gate between them. No cost share is requested, and no program income is anticipated in either phase.

4 Budget by Phase, and the Partner Site

Moore's Cancer Center is the intended partner of choice, approached at the feasibility stage according to Figure 3: a nonconfidential overview, a joint feasibility meeting with pancreatic surgery, gastrointestinal medical oncology, and trial operations, then an investigator-initiated intake with a budget and accrual estimate. UC San Diego is not a partner, sponsor, site, or endorser, because no written authorization exists.

Scope of Claims

Claimed: a demonstrated cost and schedule per work product roughly one thirtieth of the industry benchmark, a published protocol and IND, and a governance package that is site-portable by construction. Not claimed: that the simulations validate RASolute 302, that the advisory system holds any regulatory clearance, that anything here has been used in a human, or that UC San Diego has agreed to anything. Trial does not exist until a site, an IRB, an active IND, monitoring, insurance, and sponsor of record are in place.

Availability and Conflicts

Every work cited to a DOI is openly deposited; code, verification notebooks, and figure sources are in the repository [8]. The applicant holds no equity in Revolution Medicines or any robotic surgery vendor, and no sponsor, contract research organization, or medical society reviewed this document.

References

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